

PAPER_1322 — UHE Cosmic Ray Acceleration to 10^{20} eV

Author: Daniel T. Murphy **Framework:** UQFF — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology Open (CLOSED) **Date:** June 11, 2026 **Calculator:** `calculate_paradox({"paradox": "uhecr_acceleration"})`

Master Expression

$$E_{\text{max}} \approx K_{\text{MEX}} \times A_5 \times D_{\text{BSFG}} \times m_p \times c^2 \times 10^9 = 7 \times 10^{20} \text{ eV}$$

Match: covers Amaterasu 2.4×10^{20} eV

Interpretation

AGN jets accelerate via TRZ phase buoyancy. Auger + Amaterasu consistent.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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